MEGA and SVM_{Active}: Two Learning Algorithms for Learning Complex Image Query Concepts

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Abstract

We describe an image retrieval system that we have built on our recently developed query-concept learning algorithms, MEGA and SVM_{Active}. We show that MEGA and SVM_{Active} can learn a complex image-query concept in a small number of user iterations (usually three to four) on a large, multi-category, high-dimensional image database.

1 Overview

For multimedia search tasks, a query concept is hard to articulate, and articulation can be subjective. For instance, in an image search, it is difficult to describe a desired image using low-level features such as color, shape, and texture. Different users may perceive the same image differently; and even if an image is perceived similarly, users may use different vocabulary (i.e., different combinations of low-level features) to depict it. Furthermore, most users are not trained to specify simple query criteria using for example, SQL. It is thus both necessary (for capturing subjective concepts) and desirable (for alleviating users from specifying complex queries) to build intelligent search engines that can quickly learn users' query concepts via relevance feedback.

Unfortunately, traditional relevance feedback and learning techniques may not be suitable for online query-concept learning for at least two reasons. First, an on-line user is typically impatient and cannot be expected to provide a great deal of feedback, but traditional learning methods (e.g., decision trees and neural networks) often require a large number of training instances to learn a concept [6]. Second, almost all traditional relevance feedback methods [4] require users to provide good examples to seed a query. However, finding good seeds is the job of the search engine itself, and this circular requirement leaves the core problem— learning and understanding users' query concepts—unsolved.

In our demo, we present an image search engine that employs our query-concept learning algorithms—MEGA [2] and SVM$_{Active}$ [7]—to learn users' query concepts. We show that both MEGA and SVM$_{Active}$ can learn complex image-query concepts quickly and in a small number of user iterations. To demonstrate that MEGA and SVM$_{Active}$ are scalable in feature dimension, in dataset size, and in concept complexity, we construct our demo system with at least 50K *diversified* images. Each each image is characterized by more than 200 features.

While our system relies on indexers [5, 1] and perception-based image characterization [3] to be fast, robust, and scalable, the focus of this demo description is to show that our learning algorithms can quickly grasp a complex query concept with a small number of training examples. In the rest of this demo description, we briefly depict in Section 2 how MEGA and SVM$_{Active}$ work, and we present in Section 3 query scenarios and flows conducted on our system.

2 Query-Concept Learning Algorithms

While we describe MEGA and SVM$_{Active}$ in detail in [2] and [7], respectively, we highlight these two learning algorithms in this section.

2.1 MEGA

The *Maximizing Expected Generalization Algorithm* (MEGA) models query concepts in k-CNF, which can model virtually all practical query concepts. To ensure that target concepts can be learned quickly and with a small number of samples, MEGA employs two methods: sample selection (S-step) and feature reduction (F-step).

1: In its S-step, MEGA uses a k-DNF to bound the sampling space and judiciously selects samples that can collect maximum information from users to remove irrelevant k-CNF and k-DNF terms in its subsequent F-step.

2: In its F-step, MEGA removes irrelevant terms from the query-concept (i.e., a k-CNF), and at the same time, refines the sampling boundary (i.e., a k-DNF) so that the most informative samples can be selected in its subsequent S-step.

Unlike traditional query refinement methods, which use only the S-step or only the F-step, MEGA uses these two steps in a complementary way to achieve fast convergence to target concepts.

Note that MEGA does not require seeding a query. MEGA presents randomly selected images as the first round of examples. Even if all of the images generated in the first round are irrelevant, MEGA uses the negative images to reduce the set of potentially relevant images substantially (the k-DNF that bounds the sampling space can be shrunk substantially by the negative images). This the probability is high that a relevant image will be generated in the next round.

2.2 SVM$_{Active}$

SVM$_{Active}$ is an *active learning* scheme that uses support vector machines. Intuitively, SVM$_{Active}$ works by combining the following three ideas:

1. SVM$_{Active}$ regards the task of learning a target concept as one of learning an SVM binary classifier. An SVM captures the query concept by separating the relevant images from the irrelevant images with a hyperplane in a projected space, usually a very high-dimensional one. The projected points on one side of the hyperplane are considered relevant to the query concept and the rest irrelevant.

2. SVM$_{Active}$ learns the classifier **quickly** via active learning. The active part of SVM$_{Active}$ selects the most informative instances with which to train the SVM classifier. This step ensures fast convergence to the query concept in a small number of feedback rounds.

3. Once the classifier is trained, SVM$_{Active}$ returns the top-k most relevant images. These are the k images that are farthest from the hyperplane on the query concept side.

3 Illustrative Examples

We present examples in this section to show the learning steps of MEGA and SVM$_{Active}$ in two image query scenarios: *image browsing* and *similarity search*.

- Image browsing. A user knows what he/she wants but has difficulty articulating it. Through an interactive browsing session, MEGA or SVM$_{Active}$ learns what the user wants.
- Similarity search. After MEGA or SVM$_{Active}$ knows what the user wants, the search engine can perform a traditional similarity search to find data objects that appear similar to a given query object.

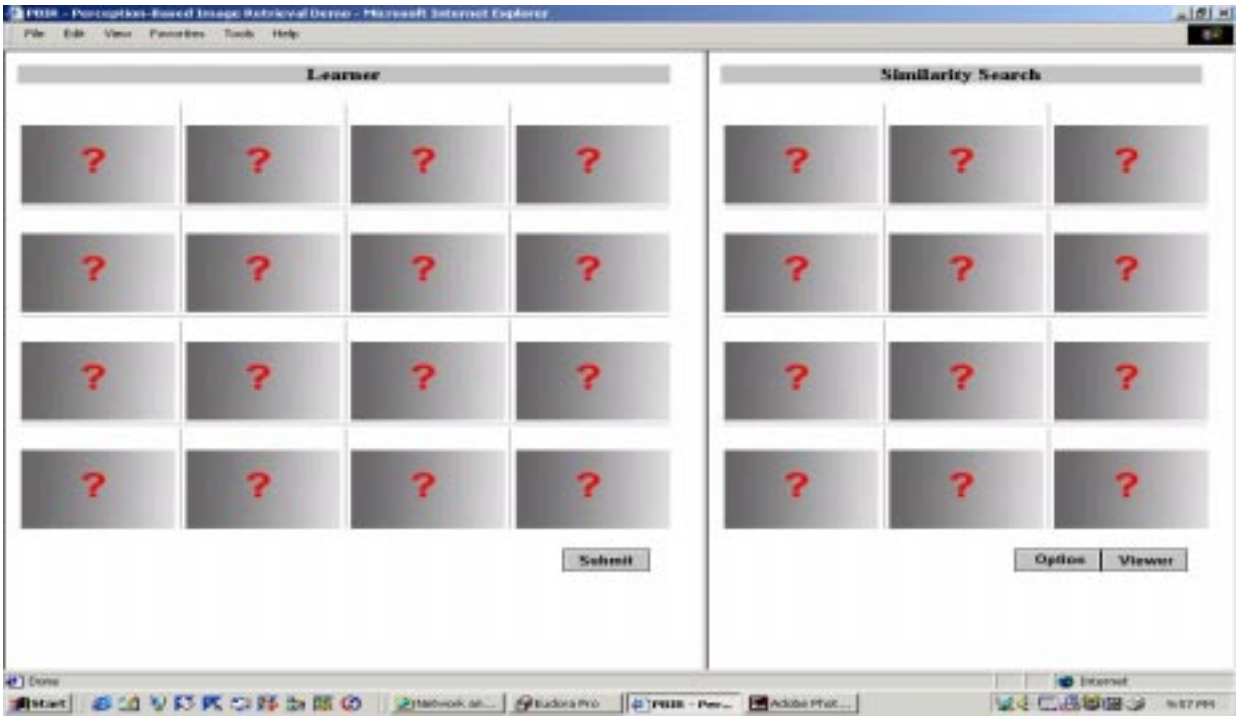


Figure 1: Wild Animal Query Screen #1.

3.1 MEGA Query Steps

In the following, we present an interactive query session using MEGA. This interactive query session involves seven screens that are illustrated in seven figures. The user's query concept in this example is "wild animals."

Screen 1. Initial Screen. Our PBIR system presents the initial screen to the user as depicted in Figure 1. The screen is split into two frames vertically. On the left-hand side of the screen is the learning frame and on the right-hand side is the similarity search frame. Through the learner, PBIR learns what the user wants via an intelligent sampling process. The similarity search frame displays what the system thinks the user wants. (The user can set the number of images to be displayed in the frames.)

Screen 2. Sampling and relevance feedback starts. Once the user clicks the "submit" button in the initial frame, the sampling and relevance feedback step commences to learn what the user wants. The PBIR system presents a number of samples in the learner frame, and the user highlights images that are relevant to his/her query concept by clicking on the relevant images.

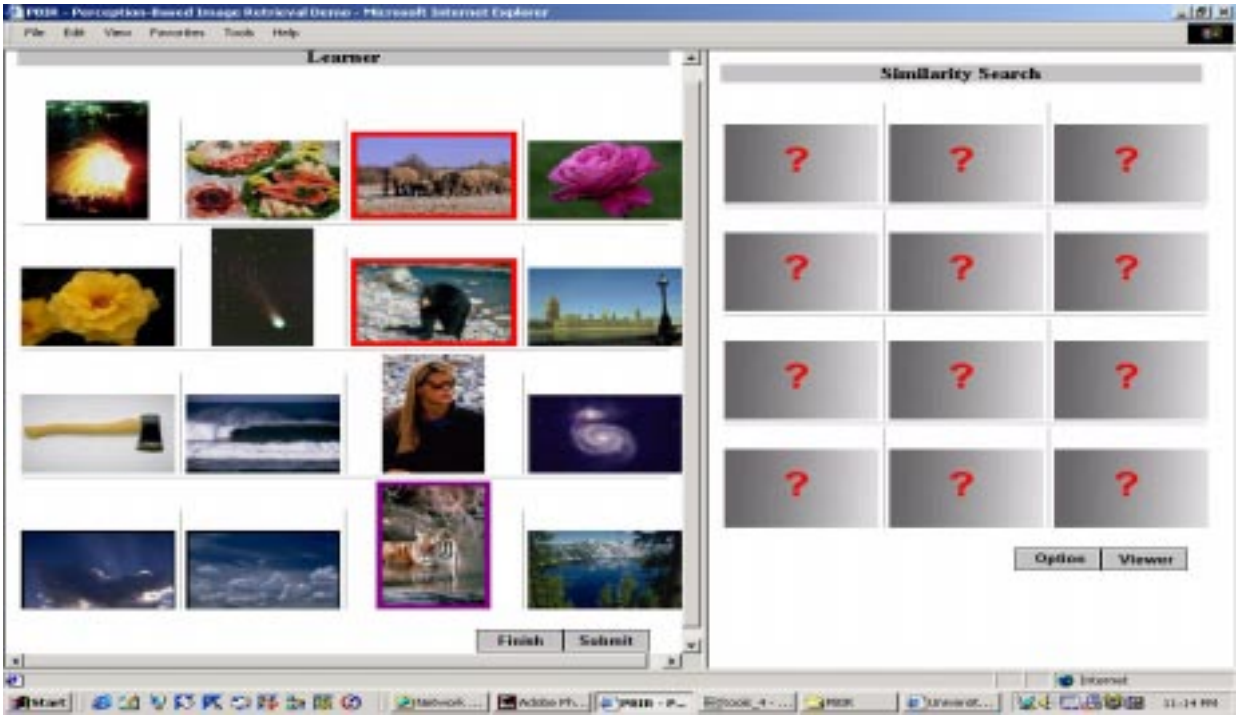


Figure 2: Wild Animal Query Screen #2.

As shown in Figure 2, two images (the third image in rows one, two and four in the learner frame) are selected as relevant, and the rest of the unmarked images are considered irrelevant. The user indicates the end of his/her selection by clicking on the submit button in the learner screen. This action brings up the next screen.

Screen 3. Sampling and relevance feedback continues. Figure 3 shows the third screen. At first, the similarity search frame does not show any image, since the system has not been able to grasp the user's query concept at this point. The PBIR system again presents samples in the learner frame to solicit feedback. The user selects the second image in the third row as the only image relevant to the query concept.

Screen 4. Sampling and relevance feedback continues. Figure 4 shows the fourth screen. First, the similarity search frame displays what the PBIR system thinks will match the user's query concept at this time. As the figure indicates, the top nine returned images fit the concept of "wild animals." The user's query concept has been captured, though somewhat fuzzily. The user can ask the system to further refine the target concept by selecting relevant images in the learner frame. In this example, the fourth image in the second row and the third image in the fourth row are selected as relevant to the concept. After the user clicks on the submit button in

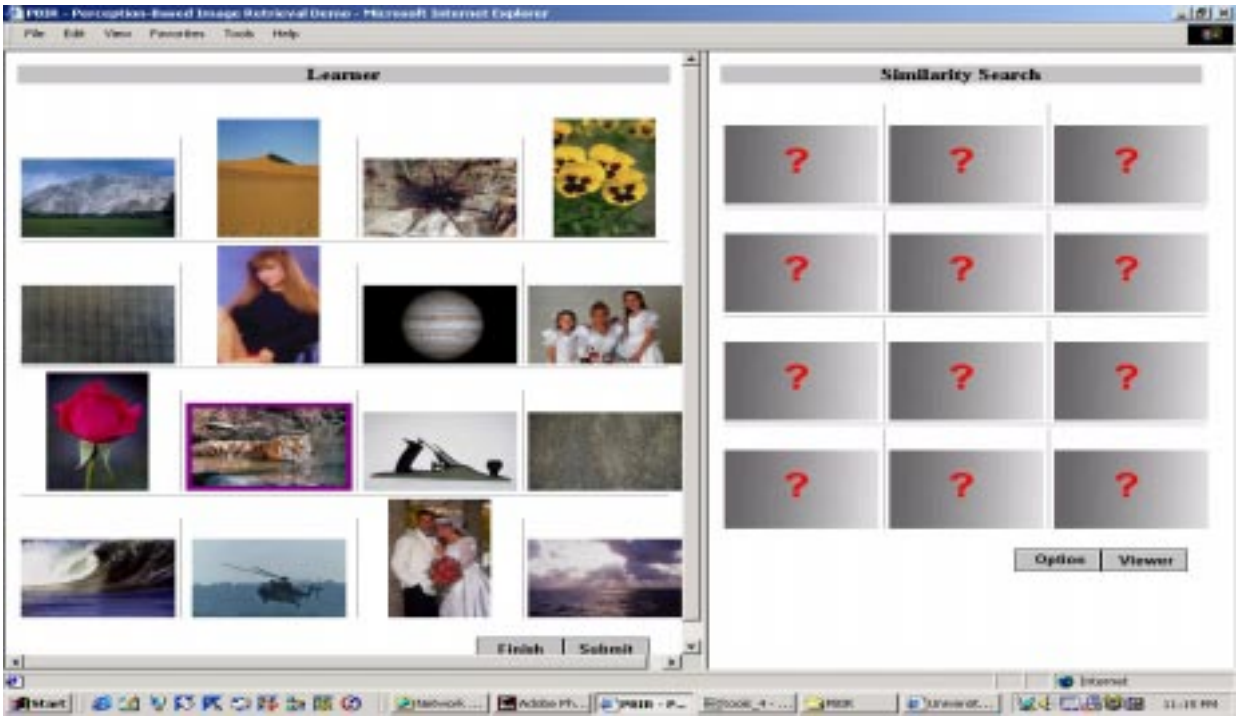


Figure 3: Wild Animal Query Screen #3.

the learner frame, the fifth screen is displayed.

Screen 5. Sampling and relevance feedback continues. The similarity search frame in Figure 5 shows that ten out of the top twelve images returned match the "wild animals" concept. The user selects four relevant images displayed in the learner frame. This leads to the final screen of this learning series.

Screen 6. Sampling and relevance feedback ends. Figure 6 shows that all returned images in the similarity search frames fit the query concept.

Screen 7. Similarity search. At any time, the user can click on an image in the similarity search frame to request images that *appear similar* to the selected images. This step allows the user to zoom in onto a specific set of images that match some appearance criteria, such as color distribution, textures and shapes. As shown in Figure 7, after clicking on one of the tiger images, the user will find similar tiger images returned in the similarity search frame. Notice that other wild animals are ranked lower than the matching tiger images, since the user has concetrated more on specific appearances than on general concepts.

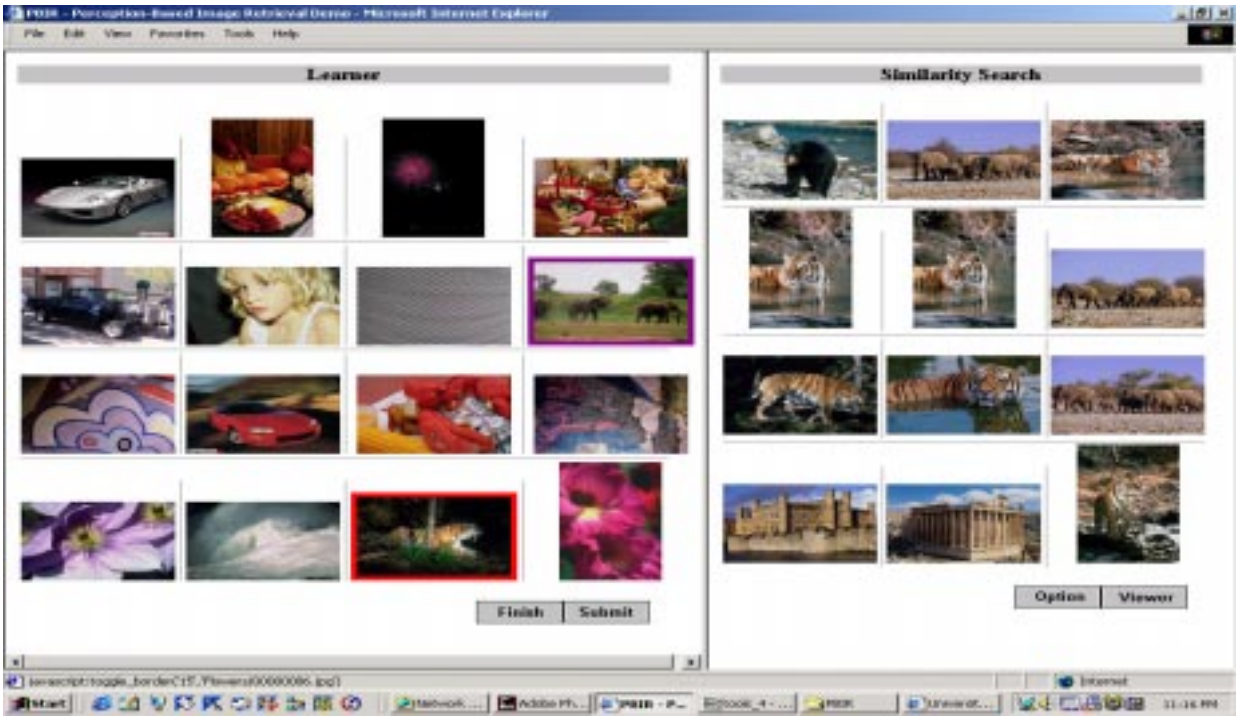


Figure 4: Wild Animal Query Screen #4.

In summary, in this example we show that our PBIR system effectively uses MEGA to learn a query concept. The images that match a concept do not have to appear similar in their low-level feature space. The learner is able to match high-level concepts to low-level features directly through an intelligent learning process. Our PBIR system can capture images that match a concept through MEGA or SVM$_{Active}$, whereas the traditional image systems can do only appearance similarity searches. As illustrated by this example, MEGA can capture the query concept of wild animal (wild animals can be elephants, tigers, bears, and etc), but a traditional similarity search engine can at best select only animals that appear similar.

In Appendix, we attach the color screen dumps of the above "wild animals" query. In addition, we attach the five query examples for five concepts: *architectures*, *fireworks*, *flowers*, *food*, and *people*. These examples show that the PBIR system can fuzzily capture a concept usually in two to three iterations and can comprehend a target concept very well in three to five iterations.

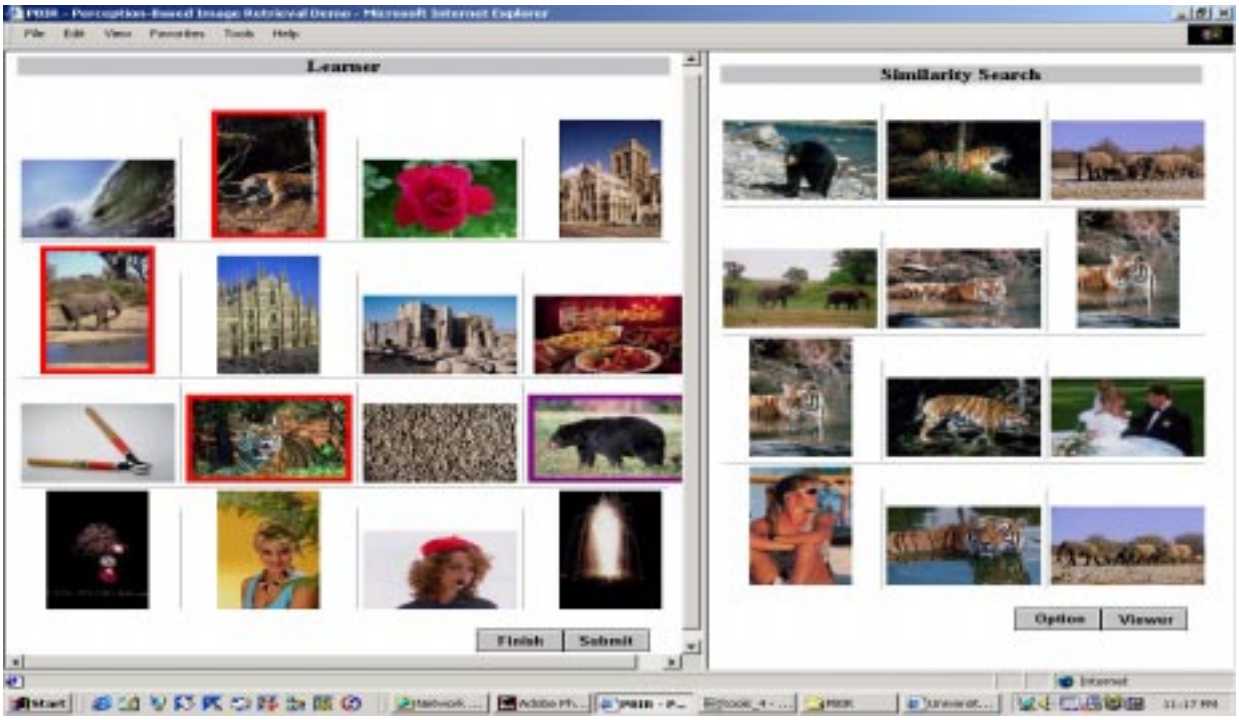


Figure 5: Wild Animal Query Screen #5.

3.2 SVM$_{Active}$ Sample Results

Finally, Figure 8 shows two sample results of using SVM$_{Active}$: one from a top-10 flowers query, and one from a top-10 tigers query. The returned images do not necessarily have the same low-level features or appearance. The returned flowers have colors of red, purple, white, and yellow, with or without leaves. The returned tiger images have tigers of different postures on different backgrounds.

4 Conclusion

Both MEGA and SVM$_{Active}$ are pioneer work for learning transient online user behavior. In an on-line learning setting, a learner is severely constrained by time and by scarcity of data, and the past access pattern of a user may not be applicable to his/her current behavior. Through our demo, we show that our learning algorithms can quickly grasp a complex query concept with a small number of labeled instances. Although we demonstrate MEGA and SVM$_{Active}$ through an image system, we believe that these learning algorithms can be applied to search for other types

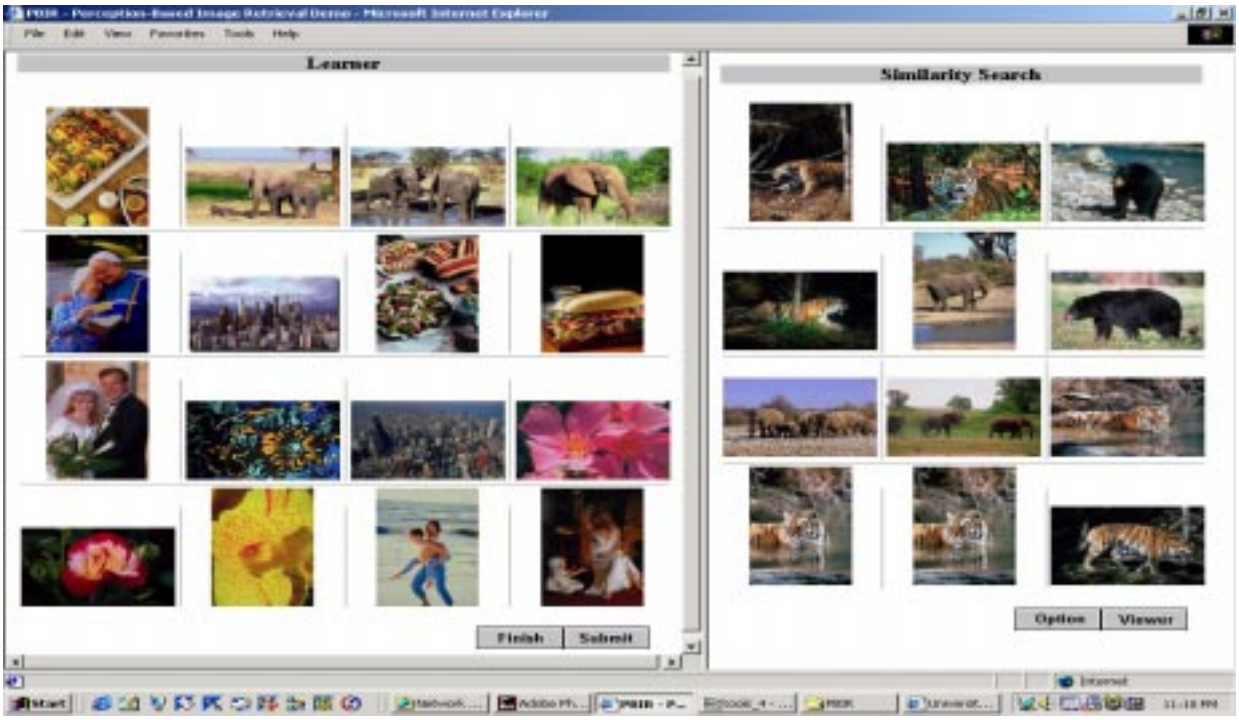


Figure 6: Wild Animal Query Screen #6.

of media data and can be used in applications such as information filtering and e-commerce.

References

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Figure 7: Wild Animal Similarity Query (Screen #7).

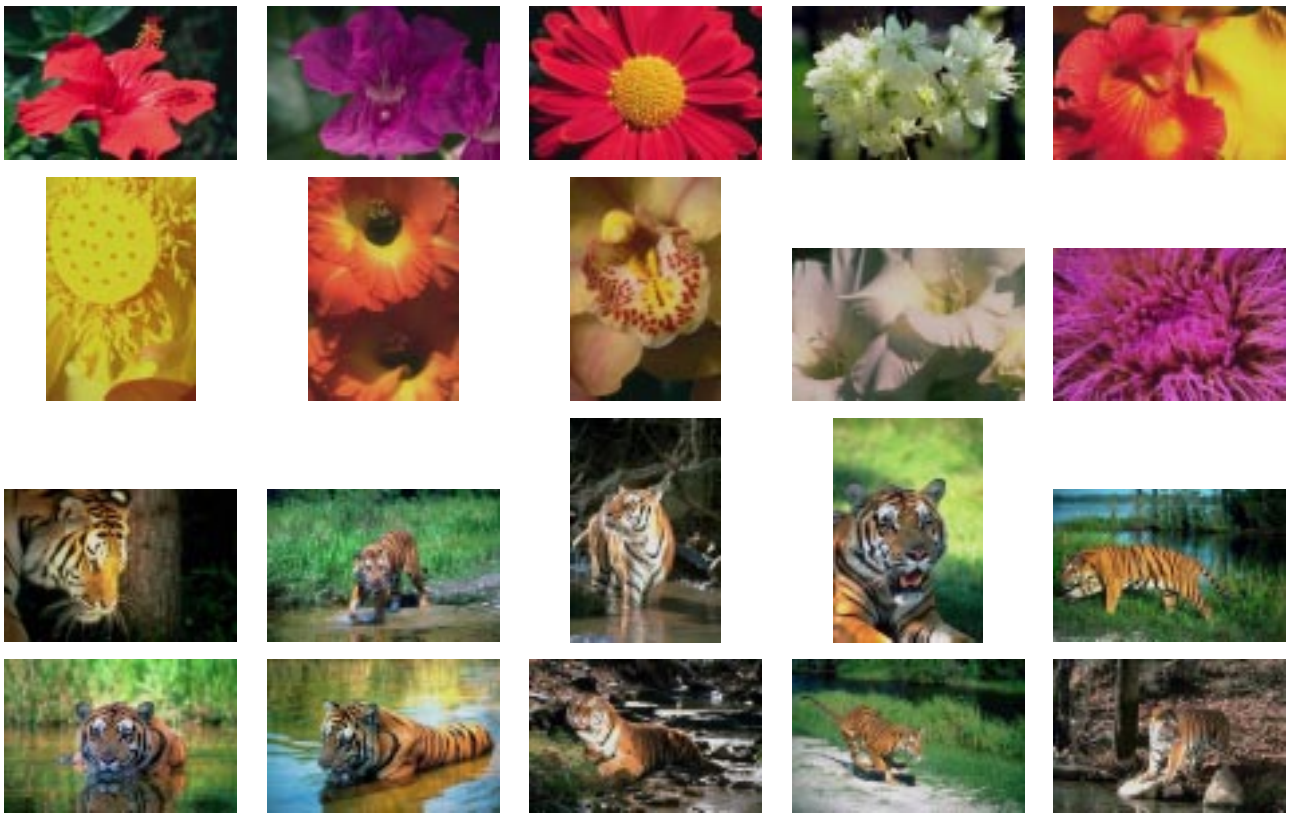


Figure 8: Flowers and Tigers Sample Query Results from SVM_{Active}.